

Linux Commands

1. Basic Linux Commands

Command	Example	Explanation
<code>pwd</code>	<code>pwd</code>	Shows the current working directory.
<code>ls</code>	<code>ls</code>	Lists files and directories.
<code>ls -l</code>	<code>ls -l</code>	Lists files with details like permissions, owner, size, and date.
<code>ls -la</code>	<code>ls -la</code>	Lists all files, including hidden files.
<code>cd</code>	<code>cd /var/log</code>	Changes directory.
<code>cd ..</code>	<code>cd ..</code>	Moves one directory back.
<code>cd ~</code>	<code>cd ~</code>	Goes to the user's home directory.
<code>clear</code>	<code>clear</code>	Clears the terminal screen.

2. File and Directory Commands

Command	Example	Explanation
<code>touch</code>	<code>touch test.txt</code>	Creates an empty file.
<code>mkdir</code>	<code>mkdir devops</code>	Creates a new directory.
<code>mkdir -p</code>	<code>mkdir -p app/logs/backup</code>	Creates parent directories if they do not exist.
<code>cp</code>	<code>cp file1.txt file2.txt</code>	Copies a file.
<code>cp -r</code>	<code>cp -r app backup-app</code>	Copies a directory recursively.
<code>mv</code>	<code>mv old.txt new.txt</code>	Renames a file.
<code>mv</code>	<code>mv app.log /tmp/</code>	Moves a file to another location.
<code>rm</code>	<code>rm test.txt</code>	Deletes a file.
<code>rm -r</code>	<code>rm -r old-folder</code>	Deletes a directory recursively.
<code>rm -rf</code>	<code>rm -rf old-folder</code>	Forcefully deletes a directory. Use carefully.

Command	Example	Explanation
<code>cat</code>	<code>cat app.log</code>	Displays file content.
<code>less</code>	<code>less /var/log/syslog</code>	Opens large files page by page.
<code>head</code>	<code>head app.log</code>	Shows the first 10 lines of a file.
<code>head -n</code>	<code>head -n 20 app.log</code>	Shows the first 20 lines.
<code>tail</code>	<code>tail app.log</code>	Shows the last 10 lines of a file.
<code>tail -f</code>	<code>tail -f app.log</code>	Shows live log updates.
<code>tail -n</code>	<code>tail -n 100 app.log</code>	Shows the last 100 lines.

3. Search Commands

Command	Example	Explanation
<code>grep</code>	<code>grep "error" app.log</code>	Searches for text inside a file.
<code>grep -i</code>	<code>grep -i "error" app.log</code>	Searches text without case sensitivity.
<code>grep -r</code>	<code>grep -r "database" /etc/</code>	Searches recursively inside directories.
<code>grep -n</code>	<code>grep -n "failed" app.log</code>	Shows matching line numbers.
<code>find</code>	<code>find / -name nginx.conf</code>	Finds files by name.
<code>find</code>	<code>find /var/log -name "*.log"</code>	Finds all <code>.log</code> files.
<code>find -mtime</code>	<code>find /var/log -mtime -1</code>	Finds files modified in the last 1 day.
<code>find -size</code>	<code>find / -type f -size +100M</code>	Finds files larger than 100 MB.
<code>locate</code>	<code>locate nginx.conf</code>	Quickly finds files by name.
<code>updatedb</code>	<code>sudo updatedb</code>	Updates the locate command database.

4. File Permission Commands

Command	Example	Explanation
<code>chmod</code>	<code>chmod 755 script.sh</code>	Changes file permissions.
<code>chmod +x</code>	<code>chmod +x deploy.sh</code>	Gives execute permission to a script.

Command	Example	Explanation
<code>chmod 644</code>	<code>chmod 644 file.txt</code>	Owner can read/write, others can only read.
<code>chown</code>	<code>sudo chown rahul file.txt</code>	Changes file owner.
<code>chown user:group</code>	<code>sudo chown rahul:devops file.txt</code>	Changes owner and group.
<code>chown -R</code>	<code>sudo chown -R rahul:devops /app</code>	Changes ownership recursively.
<code>ls -l</code>	<code>ls -l script.sh</code>	Checks file permissions and ownership.

5. User and Group Commands

Command	Example	Explanation
<code>whoami</code>	<code>whoami</code>	Shows the current logged-in user.
<code>id</code>	<code>id</code>	Shows user ID and group details.
<code>useradd</code>	<code>sudo useradd devuser</code>	Creates a new user.
<code>useradd -m</code>	<code>sudo useradd -m devuser</code>	Creates a user with home directory.
<code>passwd</code>	<code>sudo passwd devuser</code>	Sets or changes user password.
<code>usermod -aG</code>	<code>sudo usermod -aG sudo devuser</code>	Adds user to a group.
<code>groupadd</code>	<code>sudo groupadd devops</code>	Creates a new group.
<code>su</code>	<code>su - devuser</code>	Switches to another user.
<code>sudo</code>	<code>sudo systemctl restart nginx</code>	Runs a command with root privileges.

6. Process Management Commands

Command	Example	Explanation
<code>ps</code>	<code>ps</code>	Shows current shell processes.
<code>ps aux</code>	<code>ps aux</code>	Shows all running processes.
<code>grep</code> with <code>ps</code>	<code>ps aux grep nginx</code>	Finds a specific running process.
<code>top</code>	<code>top</code>	Shows live CPU and memory usage.

Command	Example	Explanation
<code>htop</code>	<code>htop</code>	Interactive process viewer.
<code>kill</code>	<code>kill 1234</code>	Kills a process using PID.
<code>kill -9</code>	<code>kill -9 1234</code>	Forcefully kills a process.
<code>pkill</code>	<code>pkill nginx</code>	Kills a process by name.
<code>jobs</code>	<code>jobs</code>	Shows background jobs.
<code>bg</code>	<code>bg</code>	Runs a stopped job in background.
<code>fg</code>	<code>fg</code>	Brings a background job to foreground.
<code>nohup</code>	<code>nohup python3 app.py &</code>	Runs a process even after logout.

7. System Information Commands

Command	Example	Explanation
<code>uname -a</code>	<code>uname -a</code>	Shows system and kernel information.
<code>hostname</code>	<code>hostname</code>	Shows system hostname.
<code>uptime</code>	<code>uptime</code>	Shows how long the system has been running.
<code>date</code>	<code>date</code>	Shows current date and time.
<code>cal</code>	<code>cal</code>	Shows calendar.
<code>who</code>	<code>who</code>	Shows logged-in users.
<code>w</code>	<code>w</code>	Shows logged-in users and their activity.

8. Disk and Storage Commands

Command	Example	Explanation
<code>df -h</code>	<code>df -h</code>	Shows disk usage in human-readable format.
<code>df -h /</code>	<code>df -h /</code>	Shows disk usage of root filesystem.
<code>du -sh</code>	<code>du -sh /var/log</code>	Shows total size of a directory.
<code>du -h --max-depth=1</code>	<code>du -h --max-depth=1 /var</code>	Shows size of each subdirectory.

Command	Example	Explanation
<code>lsblk</code>	<code>lsblk</code>	Shows disks and partitions.
<code>mount</code>	<code>sudo mount /dev/sdb1 /mnt</code>	Mounts a disk or partition.
<code>umount</code>	<code>sudo umount /mnt</code>	Unmounts a mounted filesystem.
<code>fdisk -l</code>	<code>sudo fdisk -l</code>	Lists disk partitions.

9. Memory and CPU Commands

Command	Example	Explanation
<code>free -h</code>	<code>free -h</code>	Shows memory usage.
<code>vmstat</code>	<code>vmstat 2</code>	Shows CPU, memory, and I/O stats every 2 seconds.
<code>lscpu</code>	<code>lscpu</code>	Shows CPU details.
<code>top</code>	<code>top</code>	Shows live CPU and memory usage.
<code>ps aux --sort=-%mem</code>	<code>ps aux --sort=-%mem head</code>	Shows top memory-consuming processes.
<code>ps aux --sort=-%cpu</code>	<code>ps aux --sort=-%cpu head</code>	Shows top CPU-consuming processes.

10. Networking Commands

Command	Example	Explanation
<code>ip addr</code>	<code>ip addr</code>	Shows IP addresses.
<code>ip a</code>	<code>ip a</code>	Short form of <code>ip addr</code> .
<code>ip route</code>	<code>ip route</code>	Shows routing table.
<code>ping</code>	<code>ping google.com</code>	Tests network connectivity.
<code>ping -c</code>	<code>ping -c 4 google.com</code>	Sends only 4 ping packets.
<code>curl</code>	<code>curl https://example.com</code>	Sends HTTP request.
<code>curl -I</code>	<code>curl -I https://example.com</code>	Shows only HTTP headers.

Command	Example	Explanation
<code>curl</code> API test	<code>curl http://localhost:8080/health</code>	Tests an application health endpoint.
<code>wget</code>	<code>wget https://example.com/file.zip</code>	Downloads a file.
<code>netstat</code>	<code>netstat -tulnp</code>	Shows listening ports and network connections.
<code>ss</code>	<code>ss -tulnp</code>	Modern replacement for <code>netstat</code> .
<code>ss</code> with grep	<code>ss -tulnp grep :80</code>	Checks which process is using port 80.
<code>telnet</code>	<code>telnet 10.0.0.5 3306</code>	Tests port connectivity.
<code>nc</code>	<code>nc -zv 10.0.0.5 3306</code>	Tests if a port is open.
<code>tracert</code>	<code>tracert google.com</code>	Shows the network path to a destination.
<code>dig</code>	<code>dig google.com</code>	Performs DNS lookup.
<code>nslookup</code>	<code>nslookup google.com</code>	Performs DNS lookup.

11. Package Management Commands

Ubuntu/Debian — `apt`

Command	Example	Explanation
<code>apt update</code>	<code>sudo apt update</code>	Updates package index.
<code>apt install</code>	<code>sudo apt install nginx</code>	Installs a package.
<code>apt remove</code>	<code>sudo apt remove nginx</code>	Removes a package.
<code>apt upgrade</code>	<code>sudo apt upgrade</code>	Upgrades installed packages.

RHEL/CentOS/Amazon Linux — `yum`

Command	Example	Explanation
<code>yum install</code>	<code>sudo yum install nginx</code>	Installs a package.
<code>yum remove</code>	<code>sudo yum remove nginx</code>	Removes a package.

Command	Example	Explanation
<code>yum update</code>	<code>sudo yum update</code>	Updates packages.

Fedora/RHEL 8+ — **dnf**

Command	Example	Explanation
<code>dnf install</code>	<code>sudo dnf install nginx</code>	Installs a package.
<code>dnf update</code>	<code>sudo dnf update</code>	Updates packages.
<code>dnf remove</code>	<code>sudo dnf remove nginx</code>	Removes a package.

12. Service Management Commands

Command	Example	Explanation
<code>systemctl start</code>	<code>sudo systemctl start nginx</code>	Starts a service.
<code>systemctl stop</code>	<code>sudo systemctl stop nginx</code>	Stops a service.
<code>systemctl restart</code>	<code>sudo systemctl restart nginx</code>	Restarts a service.
<code>systemctl status</code>	<code>sudo systemctl status nginx</code>	Checks service status.
<code>systemctl enable</code>	<code>sudo systemctl enable nginx</code>	Enables service on boot.
<code>systemctl disable</code>	<code>sudo systemctl disable nginx</code>	Disables service from starting on boot.
<code>journalctl</code>	<code>journalctl</code>	Shows systemd logs.
<code>journalctl -u</code>	<code>journalctl -u nginx</code>	Shows logs for a specific service.
<code>journalctl -u -f</code>	<code>journalctl -u nginx -f</code>	Follows live service logs.
<code>journalctl --since</code>	<code>journalctl --since today</code>	Shows logs from today.

13. Archive and Compression Commands

Command	Example	Explanation
<code>tar -cvf</code>	<code>tar -cvf backup.tar /app</code>	Creates a tar archive.
<code>tar -xvf</code>	<code>tar -xvf backup.tar</code>	Extracts a tar archive.
<code>tar -czvf</code>	<code>tar -czvf backup.tar.gz /app</code>	Creates compressed tar archive.
<code>tar -xzvf</code>	<code>tar -xzvf backup.tar.gz</code>	Extracts compressed tar archive.
<code>zip</code>	<code>zip app.zip app.txt</code>	Compresses a file into zip format.
<code>zip -r</code>	<code>zip -r app.zip /app</code>	Compresses a directory.
<code>unzip</code>	<code>unzip app.zip</code>	Extracts a zip file.

14. Text Processing Commands

Command	Example	Explanation
<code>awk</code>	<code>awk '{print \$1}' access.log</code>	Prints the first column from a file.
<code>awk</code>	<code>awk '{print \$1, \$3}' file.txt</code>	Prints first and third columns.
<code>awk</code> with <code>df</code>	<code>df -h awk '{print \$1, \$5}'</code>	Prints filesystem and usage percentage.
<code>sed</code>	<code>sed 's/dev/prod/g' config.txt</code>	Replaces <code>dev</code> with <code>prod</code> in output.
<code>sed -i</code>	<code>sed -i 's/dev/prod/g' config.txt</code>	Replaces text directly inside the file.
<code>cut</code>	<code>cut -d ':' -f1 /etc/passwd</code>	Cuts and prints specific fields.
<code>sort</code>	<code>sort names.txt</code>	Sorts lines alphabetically.
<code>sort -r</code>	<code>sort -r names.txt</code>	Sorts lines in reverse order.
<code>uniq</code>	<code>sort names.txt uniq</code>	Removes duplicate lines.
<code>uniq -c</code>	<code>sort names.txt uniq -c</code>	Counts duplicate lines.
<code>wc</code>	<code>wc file.txt</code>	Counts lines, words, and characters.
<code>wc -l</code>	<code>wc -l file.txt</code>	Counts only lines.

15. Redirection and Pipe Commands

Command	Example	Explanation
<code>></code>	<code>echo "hello" > file.txt</code>	Writes output to a file and overwrites existing content.
<code>>></code>	<code>echo "new line" >> file.txt</code>	Appends output to a file.
<code> </code>	<code>cat app.log grep error</code>	Sends output of one command to another.
<code>2></code>	<code>ls /wrongpath 2> error.log</code>	Redirects errors to a file.
<code>&></code>	<code>command &> output.log</code>	Redirects both output and error to a file.

16. Environment Variable Commands

Command	Example	Explanation
<code>env</code>	<code>env</code>	Shows all environment variables.
<code>echo</code>	<code>echo \$PATH</code>	Prints the value of a variable.
<code>export</code>	<code>export ENV=dev</code>	Creates an environment variable.
<code>export</code>	<code>export DB_HOST=localhost</code>	Sets application-related environment variable.
<code>unset</code>	<code>unset DB_HOST</code>	Removes an environment variable.

17. SSH and Remote Server Commands

Command	Example	Explanation
<code>ssh</code>	<code>ssh user@server-ip</code>	Connects to a remote server.
<code>ssh</code>	<code>ssh ubuntu@10.0.0.5</code>	Connects to server using username and IP.
<code>ssh -i</code>	<code>ssh -i key.pem ubuntu@10.0.0.5</code>	Connects using a private key.
<code>scp</code>	<code>scp app.log ubuntu@10.0.0.5:/tmp/</code>	Copies file from local to remote server.

Command	Example	Explanation
<code>scp</code>	<code>scp ubuntu@10.0.0.5:/tmp/app.log .</code>	Copies file from remote to local machine.
<code>scp -i</code>	<code>scp -i key.pem app.log ubuntu@10.0.0.5:/tmp/</code>	Copies file using private key.
<code>rsync</code>	<code>rsync -av /app/ ubuntu@10.0.0.5:/backup/app/</code>	Syncs files efficiently.
<code>rsync -e</code>	<code>rsync -av -e "ssh -i key.pem" /app/ ubuntu@10.0.0.5:/backup/app/</code>	Syncs files using SSH private key.

18. Log Troubleshooting Commands

Command	Example	Explanation
<code>tail -f</code>	<code>sudo tail -f /var/log/syslog</code>	Checks live system logs on Ubuntu.
<code>tail -f</code>	<code>sudo tail -f /var/log/auth.log</code>	Checks authentication logs on Ubuntu.
<code>tail -f</code>	<code>sudo tail -f /var/log/messages</code>	Checks system logs on RHEL/CentOS.
<code>grep</code>	<code>grep "Failed password" /var/log/auth.log</code>	Checks failed login attempts.
<code>journalctl -u</code>	<code>journalctl -u nginx -f</code>	Checks live logs of a service.
<code>tail -f</code>	<code>tail -f /app/logs/app.log</code>	Checks live application logs.
<code>dmesg</code>	<code>dmesg</code>	Shows kernel logs.
<code>dmesg -w</code>	<code>dmesg -w</code>	Follows live kernel logs.

19. Cron Job Commands

Command	Example	Explanation
<code>crontab -e</code>	<code>crontab -e</code>	Edits cron jobs.
<code>crontab -l</code>	<code>crontab -l</code>	Lists current cron jobs.

Command	Example	Explanation
Cron every 5 minutes	<code>* /5 * * * * /home/ubuntu/healthcheck.sh</code>	Runs script every 5 minutes.
Cron daily at 2 AM	<code>0 2 * * * /backup.sh</code>	Runs script daily at 2 AM.
Cron every Sunday	<code>0 1 * * 0 /cleanup.sh</code>	Runs script every Sunday at 1 AM.

Cron format:

Field	Meaning
Minute	<code>0-59</code>
Hour	<code>0-23</code>
Day of month	<code>1-31</code>
Month	<code>1-12</code>
Day of week	<code>0-7</code> , where <code>0</code> and <code>7</code> mean Sunday
Command	Script or command to execute

20. DevOps Troubleshooting Commands

Task	Command Example	Explanation
Check service status	<code>systemctl status nginx</code>	Checks whether nginx is running or failed.
Restart service	<code>sudo systemctl restart nginx</code>	Restarts a service.
Check open ports	<code>ss -tulnp</code>	Shows listening ports.
Check CPU usage	<code>top</code>	Shows live CPU usage.
Check memory	<code>free -h</code>	Shows memory usage.
Check disk usage	<code>df -h</code>	Shows filesystem disk usage.
Check directory size	<code>du -sh /var/log</code>	Shows size of <code>/var/log</code> .
Check live logs	<code>tail -f /var/log/syslog</code>	Follows live system logs.
Find process	<code>ps aux grep java</code>	Finds Java process.
Kill process	<code>kill -9 <PID></code>	Forcefully kills a process.

Task	Command Example	Explanation
Test API	<code>curl http://localhost:8080/health</code>	Checks application health endpoint.
Check port usage	<code>sudo lsof -i :8080</code>	Finds process using port 8080.
Check reboot time	<code>who -b</code>	Shows last system boot time.
Check login history	<code>last</code>	Shows user login history.
Watch command output	<code>watch df -h</code>	Runs <code>df -h</code> repeatedly every 2 seconds.

21. Important One-Liners

Command Example	Explanation
<code>find / -type f -exec du -h {} + 2>/dev/null sort -rh head -n 10</code>	Finds top 10 largest files.
<code>ps aux --sort=-%mem head</code>	Shows top memory-consuming processes.
<code>ps aux --sort=-%cpu head</code>	Shows top CPU-consuming processes.
<code>sudo ss -ltnp</code>	Shows all listening TCP ports.
<code>ls -l wc -l</code>	Counts number of files in current directory.
<code>grep -i "error" /var/log/syslog</code>	Searches error messages in system logs.
<code>tar -czvf logs-backup.tar.gz /var/log</code>	Compresses log directory.
<code>find /var/log -name "*.log" -mtime +7</code>	Finds log files older than 7 days.
<code>du -ah /var/log sort -rh head -n 10</code>	Shows largest files/directories inside <code>/var/log</code> .

22. Most Important Linux Commands for Interviews

Command	Example	Explanation
<code>ls -la</code>	<code>ls -la</code>	Lists all files with details.
<code>cd</code>	<code>cd /etc</code>	Changes directory.
<code>pwd</code>	<code>pwd</code>	Shows current directory.
<code>cat</code>	<code>cat file.txt</code>	Displays file content.
<code>less</code>	<code>less app.log</code>	Opens large files page by page.
<code>head</code>	<code>head app.log</code>	Shows first 10 lines.
<code>tail -f</code>	<code>tail -f app.log</code>	Follows live logs.
<code>grep</code>	<code>grep error app.log</code>	Searches text in files.
<code>find</code>	<code>find / -name file.txt</code>	Finds files.
<code>chmod</code>	<code>chmod +x script.sh</code>	Changes permissions.
<code>chown</code>	<code>chown user:group file.txt</code>	Changes ownership.
<code>ps aux</code>	<code>ps aux</code>	Shows running processes.
<code>top</code>	<code>top</code>	Shows live system usage.
<code>kill</code>	<code>kill -9 PID</code>	Kills a process.
<code>df -h</code>	<code>df -h</code>	Checks disk usage.
<code>du -sh</code>	<code>du -sh /var/log</code>	Checks directory size.
<code>free -h</code>	<code>free -h</code>	Checks memory.
<code>ip a</code>	<code>ip a</code>	Shows IP address.
<code>ping</code>	<code>ping google.com</code>	Checks connectivity.
<code>curl</code>	<code>curl http://localhost:8080/health</code>	Tests HTTP/API endpoint.
<code>ss -tulnp</code>	<code>ss -tulnp</code>	Checks open ports.
<code>systemctl status</code>	<code>systemctl status nginx</code>	Checks service status.
<code>journalctl -u</code>	<code>journalctl -u nginx -f</code>	Checks service logs.
<code>ssh</code>	<code>ssh ubuntu@server-ip</code>	Connects to remote server.
<code>scp</code>	<code>scp file.txt ubuntu@server:/tmp/</code>	Copies files between systems.
<code>tar</code>	<code>tar -czvf backup.tar.gz /app</code>	Compresses files/directories.
<code>crontab -e</code>	<code>crontab -e</code>	Edits scheduled jobs.

23. Interview Scenario-Based Linux Commands

Scenario	Command Example	Explanation
Application is down	<code>systemctl status app-service</code>	Check whether service is running or failed.
Need to check app logs	<code>journalctl -u app-service -f</code>	View live service logs.
Disk is full	<code>df -h</code>	Check filesystem disk usage.
Find large files	<code>du -ah / sort -rh head -n 10</code>	Finds biggest files/directories.
High CPU issue	<code>top</code>	Check which process is consuming CPU.
High memory issue	<code>ps aux --sort=-%mem head</code>	Finds top memory-consuming processes.
Port conflict	<code>sudo lsof -i :8080</code>	Finds which process is using port 8080.
Network issue	<code>ping google.com</code>	Checks basic connectivity.
DNS issue	<code>nslookup google.com</code>	Checks DNS resolution.
API not responding	<code>curl -I http://localhost:8080</code>	Checks HTTP response headers.
Permission issue	<code>ls -l file.txt</code>	Checks file permission.
Fix script permission	<code>chmod +x deploy.sh</code>	Adds execute permission.
SSH issue	<code>ssh -v user@server-ip</code>	Debugs SSH connection.
Check failed login	<code>grep "Failed password" /var/log/auth.log</code>	Finds failed SSH login attempts.
Check recent logs	<code>journalctl --since "1 hour ago"</code>	Shows logs from the last hour.